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An Open, Flexible Framework for Binary Orbital Parameter Inference: Modeling Spectroscopic Eclipsing Binaries with APOGEE and TESS data

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ABSTRACT

Binary-star systems play an essential role in addressing a wide array of astrophysical problems, from developing accurate distance ladders to calibrating stellar evolution models. Thanks to wide-field time-domain spectroscopic and photometric surveys, we now have the capacity to construct catalogs of binary-star orbital parameters across the entire sky using both radial velocity and photometric time series data. We present a data-driven probabilistic framework to infer stellar and orbital parameters for detached single-linedspectroscopic eclipsing binary-star systems (SEBs).

The method employs a modern gradient-based inference and is generalized to require as little manual intervention as possible. We test this framework on a sample of 12 SEBs with well-sampled radial-velocity phase coverage in the APOGEE survey and varied eclipse photometry from TESS. Our framework successfully generates full orbital parameter posterior constraints for 11 binary-star systems in our APOGEE–TESS sample and partial orbital parameter posterior constraints for one binary-star system. We report the properties of some note-worthy individual binary-star systems within our sample, discuss the release of our framework as an open-source package for use by the astronomical community, and propose future development of a benchmark catalog of SEB stellar parameters via application of our framework applied to a much larger sample of systems in the APOGEE–TESS overlap region with fewer radial-velocity measurements.

1. INTRODUCTION

Binary-star systems (hereafter "binaries") are a fundamental component of a plethora of astrophysical problems and are very common throughout the universe. About half of the stars in the Milky Way are found in binaries (Duchene et al. 2018), and the binary demographics and evolution of binaries are key unknowns in many astrophysical contexts (e.g., Price-Whelan et al. 2019; Rix et al. 2019). However, binaries are also critical benchmark systems: for example, binaries provide one of the few ways to infer stellar distances without the use of stellar evolution models (Soderblom 2010; Serenelli et al. 2021), and enable an important channel for calibrating models of stellar parameters

across a range of stellar masses (Kraus & Hillenbrand 2009; Stacy et al. 2009) as most binary-star components are assumed to be coeval in the field.

A crucial class of benchmark binaries are eclipsing binaries (Serenelli et al. 2021) (EBs), in which the eclipse shapes and timing provide information about both the orbital properties of the system, such as inclination, period, eccentricity, and orientation, and also intrinsic properties of the stellar components, such as the radius ratio and surface brightness ratio (e.g. Andersen 1991; Torres et al. 2010a; Miller et al. 2020; Cunningham et al. 2020). Especially useful EB systems are those with small secondary components (e.g., Torres 2013; Gill et al. 2020; Sebastian et al. 2022) in which a star with well-predicted structure and parameters (e.g., a 1 $M_\odot$ primary star) has a low-mass companion (e.g., a 0.3 $M_\odot$ star) with much more uncertain intrinsic and atmospheric parameters (Kraus et al. 2011; Cassisi & Salaris 2019). Spectroscopic binaries, on the other hand, can offer different constraints on orbital parameters depending on whether the spectra encodes a detection of one (SB1) or two (SB2) stars. However, due to a degeneracy between the inclination and semi-major axis, SBs alone can only provide population level inferences of binary demographics (Badenes et al. 2018; Price-Whelan et al. 2020). SB1s, coupled with available ancillary data, are ideal to perform analyses of binary-star field and cluster populations.

Both historically and continuing today, benchmark EB systems are often first identified from time-domain photometric observations and subsequently observed spectroscopically to obtain radial velocity time series of the systems (e.g. Baroch et al. 2022; Hong et al. 2022). With recent all-sky, high-cadence, time-domain photometric surveys such as the Kepler Mission (Prša et al. 2011) and the Transiting Exoplanet Survey Satellite (TESS; Ricker et al. 2014; Stassun et al. 2018), it has become possible to identify large samples of $\sim 10^4$ EB systems across a range of stellar types (Kirk et al. 2016; Prša et al. 2022). These systems then require spectroscopic follow-up to obtain radial velocity and spectroscopic stellar parameter constraints to determine accurate intrinsic parameters (masses, radii, etc.), but follow-up programs at this scale are challenging given the required time-domain observations needed. Luckily, large spectroscopic surveys, in terms of the number of targets, obtain time-resolved spectroscopic data as a part of their survey strategy or targeting programs, such as the APO Galactic Evolution Experiment (APOGEE; Majewski et al. 2017).

In the case of APOGEE, the time coverage is typically sparse and over long baselines (months to a decade; see, e.g., Figure 2 in Price-Whelan et al. 2020). Still, APOGEE observed over 50,000 observed sources have more than 5 spectroscopic epochs. In this work, we are ultimately interested in generating a framework that can reliably infer orbital parameters for these EBs with sparse or small number of radial velocity measurements. Radial velocities are difficult and expensive to measure with high enough cadences and over time baselines *en masse*. Mining the population of EBs with sparse time-series radial velocity data for orbital parameters will enlarge the overall sample of binary parameters from which population demographics can be better constrained.

Here we develop a framework for the generalized joint modeling of SEBs with this goal in mind; we use a smaller subset of SEBs as a testbed to demonstrate the method, characterize its reliability and precision, and understand its limitations.

Multiple packages already exist and are currently in use by the community to perform joint fitting to time-series radial velocities and light curve data, such as PlanetPack (Baluev 2013; Baluev et al. 2020), Allesfitter (Günther & Daylan 2021), Exofast (Eastman et al. 2013, 2019), Exoplanet (Foreman-Mackey et al. 2021), Juliet (Espinoza et al. 2019), and Ellc (Maxted 2016). However, a majority of

these packages are typically well-suited to fit to double-lined spectroscopic binaries (SB2s) or light curves with multiple eclipsing exoplanets. These modeling packages employ Bayesian inference to obtain orbital solutions and intrinsic parameters of the component orbiting bodies based on the observed data provided. The utility of these packages is that they enable constructing or specifying a likelihood function for the observed data simultaneously given parameters that pertain to both the intrinsic stellar parameters of the binary components and the orbital architecture of a system. This likelihood function can then be combined with prior probability distribution functions (PDFs) to construct a posterior PDF for the parameters.

Modern modeling frameworks usually perform Markov Chain Monte Carlo (hereafter MCMC) sampling to draw samples from the posterior PDF, which can then be used to compute statistics of the posterior or marginal posteriors for parameters of interest. Different packages offer different methods to perform MCMC sampling (such as ensemble sampling methods, nested sampling, or Hamiltonian Monte Carlo methods), each with advantages and disadvantages for generating posterior samples. For example, ensemble sampling methods typically have trouble generating posterior samples when the shape of the posterior PDF is highly-correlated or irregular, and generally do not scale well to large numbers of parameters. Nested sampling handles multimodality and complex posterior PDF shapes better, but also scales poorly with the number of parameters. Hamiltonian Monte Carlo (hereafter HMC) methods can generally handle more irregular posterior PDF shapes and scale well with the number of parameters. HMC performs inference by using the gradients of the posterior PDF with respect to the parameters to explore it more efficiently. However, calculating the gradients during sampling can be computationally expensive.

The existing EB modeling frameworks currently implement one or more of these sampling methods, which we briely discuss. Allesfitter (Günther & Daylan 2021), for example, uses ensemble or nested sampling methods through the emcee (Foreman-Mackey et al. 2013) or dynesty packages (Speagle 2020), respectively. Juliet (Espinoza et al. 2019) enables modeling time-series SB1 data with light curve data using nested sampling. Exoplanet (Foreman-Mackey et al. 2021) can fit both flux and radial velocity time-series data and is able to use HMC sampling methods because it is built upon the PyMC3 (Salvatier et al. 2016) model building framework which is optimized for fast gradient calculations. This infrastructure is built with performing fast calculations of gradients in probabilistic models in mind, correcting a computational bottleneck when performing HMC sampling.

One of the main issues with constructing a flexible, generalized Bayesian probabilistic framework is the necessity to be able to handle many different and highly-irregular posterior parameter spaces. The data we want to model, light curve flux measurements and time-series radial velocity measurements, are both influenced by the orbital parameters of their host binary system, as well as the orientation in space with respect to us. Jointly modeling both the radial velocity data with the light curve allows us to infer the binaries orbital parameters and orientation completely. However, the posterior PDF can also be irregularly-shaped due to choices of parameterization or complex degeneracies between parameters, causing many samplers to under-perform or fail.

The implementation of HMC used by Exoplanet includes the No U-Turn Sampler (hereafter NUTS, see Hoffman & Gelman 2014). NUTS is a modification of an HMC sampler that tunes the model's hyperparameters as samples are drawn to efficiently explore the posterior parameter space. Tuning hyperparameters otherwise typically involves costly re-runs of models with different hyperparameter values.

We build our framework using Exoplanet as we focus on flexibility to be able to properly infer orbital parameters for a varied combination of SEBs with only SB1 data and automation to tackle a large enough sample to infer binary population demographics. The NUTS algorithm to perform flexible HMC inference without costly external hyperparameter tuning meets both design requirements of our proposed framework.

We demonstrate the capabilities of this framework with a sample of spectroscopic eclipsing binaries with light curve data from the TESS mission and single-line radial velocity data from the APOGEE surveys. Currently, there has not yet been an investigation into the capabilities of jointly fitting SEBs with TESS and APOGEE data en masse. We hope to showcase both the exquisite nature of the high quality data within TESS and APOGEE, as well as the robustness of our framework as a tool to the astronomical community.

2. DATA

The successor to the Kepler spacecraft (Batalha et al. 2010), the Transit Exoplanet Survey Satellite (hereafter, TESS) (Ricker et al. 2014; Sullivan et al. 2015; Stassun et al. 2017, 2019) is now in its 5th year of operation in the second extended mission. The time-domain photometric data from TESS has helped make breakthroughs in the field of exoplanets and in the study of binaries because of its high-cadence, nearly all-sky-coverage observations of over 200,000 stars, covering almost 70% of the night sky. TESS observes targets continuously within a particular sector over a span of 13.5 with a short 2-minute cadence; each sector has at least 27 total days of observing time.

The resulting data contains a large number of high quality time-series light curve flux data for each star in the TESS Input Catalog (Stassun et al. 2017). Having this continuous, short-cadence light curve data enables studying periodic behavior present in the observed stars over many timescales (e.g., stellar flares (Günther & Daylan 2021), stellar rotation (Doyle et al. 2020), stellar pulsations (Cunha et al. 2019), and stellar/sub-stellar companions (Prša et al. 2022; Gandolfi et al. 2018, respectively).

The Sloan Digital Sky Surveys (SDSS) have been in operation for nearly 20 years. The recently released 17th data release from SDSS (Abdurro'uf et al. 2022) marks the final data release in SDSS-IV (Blanton et al. 2017), and contains a full release of the APOGEE-2 spectroscopic data (spectral resolution $R \sim 22,500$) (Majewski et al. 2017; Beaton et al. 2021; Santana et al. 2021). The APOGEE-2 data set contains over ~ 2.6 million individual spectra for over $700,000$ stars. The APOGEE project was originally designed to only perform spectroscopic chemical analysis of stars, which required multiple visits to accumulate a high enough signal to noise ratio. With multiple spectra and long time baselines, these data can be repurposed to provide time series radial velocity measurements (e.g., Troup et al. 2016; Badenes et al. 2018; Price-Whelan et al. 2018).

For our initial list of binaries we sought to have well-sampled radial velocity data. We opted to use the candidate binaries identified by modeling the time-resolved "epoch" radial velocity data in APOGEE data release 17 (Abdurro'uf et al. 2022), as released in the *The Joker* value-added catalog. This sample was identified using the custom Monte Carlo sampler *The Joker* (Price-Whelan et al. 2017), which was designed to use radial velocity measurements from single lined binaries (SB1s) to fit a six parameter binary orbital model and infer posterior distributions with the capacity to handle multi-modal solutions. The pipeline used to process and run *The Joker* on the APOGEE DR17 data follows the procedure used in Price-Whelan et al. (2020). We will demonstrate with our framework that the inclusion of photometric data into the orbital model will break some of the degeneracies present when only modeling with radial velocities.

Further quality cuts were implemented on this catalog of binaries from APOGEE DR17 to produce a "gold sample" that can be reliably employed to consider orbital properties with stellar parameters. In general we require that each system have clean APOGEE data, reliable APOGEE stellar parameters, well-behaved posteriors from *The Joker*, and measurements determined by the STARHORSE project. We restate those quality cuts below; in particular, we require that the APOGEE binary-star systems have:

1. unimodal (in orbital period) posterior samples produced from *The Joker*,
2. close cross-matches with *Gaia* and 2MASS within ($2''$),
3. analyzed by the STARHORSE project (Queiroz et al. 2018),
4. no APOGEE fiber contaminants ($< 2''$ with $\Delta G > -5$), using *Gaia* EDR3 (Gaia Collaboration & Brown 2020) source positions and photometry,
5. no bright neighbors ($< 10''$ with $\Delta G > -2.5$), again using *Gaia* EDR3 photometry,
6. reliable stellar parameters from the APOGEE pipeline,[1]
7. small inferred excess radial-velocity variance ($s_{MAP} < 0.5$ km s^{-1}),
8. more than 5 APOGEE visits ($\mathtt{N}_{visits} > 5$).

The resulting Gold sample contains 2,666 binary systems. We truncate the gold sample further by requiring that all binary systems have a spectroscopic orbital period less than 30 days according to the maximum *a posteriori* period sample generated from *The Joker* for each system. We set this period limit to 30 days to limit our initial sample to low eccentricity binaries, due to tidal circularization of tight binaries. We reduce our binary sample to 1,741 binary systems with this period cut.

We choose to only consider pre-processed TESS light curves from the TESS-SPOC processing group (Caldwell et al. 2020). This is not a fundamental step in our framework, nor is it necessary for the proper employment of our framework, but is a choice in data to reduce the amount of pre-processing of TESS full frame images to produce the light curves used in this work. We build our framework to be able to handle any time-series data with flux measurements, time-stamps, and flux uncertainties.

We employ the LightKurve (Lightkurve Collaboration et al. 2018) Python module to search and download all available TESS short cadence (2 min) light curves across all sectors for the 1,741 spectroscopic binary systems identified from *The Joker* and APOGEE data. We find 104 binary systems with TESS light curve data. We find that there are 57 binary systems in our sample with noticeable or conspicuous eclipses visible in the light curve after a rudimentary BLS period determination using LightKurve. We visual inspect the resulting light curves for both a primary and secondary eclipse to limit our sample to non-grazing orbital orientations. Our final sample has 12 binaries which met this requirement.

3. METHODS

We construct a generalized framework that will be used to build different SEB orbital models with parameters constrained using both the radial velocity measurements from APOGEE and the light

[1] $-0.5 < \log g < 5.5$
$3500\ \mathrm{K} < T_{\mathrm{eff}} < 10\,000\ \mathrm{K}$
$-2.5 < [\mathrm{M/H}] < 0.5$

Figure 1. Each panel shows a phase-folded TESS light curve for the 12 spectroscopic eclipsing binary systems in our sample. All light curves have been phase-folded with a Box Least Squares-derived period and eclipse epoch from the TESS light curves, except for TIC 169820068, which is phase-folded using the maximum *a posteriori* period from *The Joker* using APOGEE radial velocities. The flux values for all systems have been normalized to parts per thousand, ppt, and plotted in black, and phases are shown in time units. A rolling median flux of the light curve is over-plotted as the solid (orange) line. The light curves selected for this sample have a wide range of eclipse geometries, making them suitable as an initial test sample for our framework.

curve flux measurements from TESS. To that end, we design our framework to perform a series of functions such that flexibility is maximized while only minimal external intervention is required once the framework is initiated for a target binary system.

The likelihood function in our framework comes from the internal orbital model Exoplanet assuming a Keplerian orbit. The orbital model is constructed with the primary component mass and radius, the mass ratio, the surface brightness ratio, the radius ratio, along with the eccentricity, argument of periastron, impact parameter, and period. The secondary component mass and ratio are depend on the primary component parameters and ratios mentioned above. The light curve model is constrained by these parameters as well as limb darkening parameters for the primary and secondary component.

We list the full suite of parameters that go into our framework along with short descriptions in Table 1. We initialize the framework with the period and eclipse epoch times derived in Section 2, where we perform a manual inspection to decide to utilize either the TESS period/eclipse epoch or the APOGEE *The Joker*-derived period/eclipse epoch. Future versions of our framework may rely on calculating the signal detection efficiency (SDE, Alcock & Parker 2019) of phase folded data to automatically select best period/eclipse epoch combination.

We calculate initial estimates for some of the parameters we use to initialize our priors using the Isochrones python package. We infer initial individual component stellar parameters by sampling the MIST (Dotter 2016) evolutionary isochrones built into the Isochrones package. These posterior PDFs derived from Isochrones help inform the prior PDFs used in our orbital modeling framework for the log-primary mass $\log M_1$, the log-primary radius $\log R_1$, the log-radius ratio $\log k$, the log-mass ratio $\log q$, and the log-surface brightness ratio $\log s$.

With these estimates we initialize our priors, as well as our eclipse model likelihood function, within a PyMC3 model within the Exoplanet package. We sample using HMC to construct the posterior SEB orbital parameters, allowing effective Bayesian sampling within ill-behaved and high dimensional parameter spaces. Posterior distributions which are multi-modal or have irregularly shaped geometries are an obstacle for any Monte Carlo sampler. Spectroscopic eclipsing binaries, in particular, can have high dimensional posterior parameter spaces. These issues stem from the combination of intrinsic binary dynamics as well as observational orientations from our line of sight. There are also numerous degeneracies between these two groups of parameters, further complicating sampling the posteriors effectively. HMC sampling, by design, is able to sample posteriors such as these and is the best choice to build our generalized framework.

3.1. *Why use Hamiltonian Monte Carlo?*

The most frequently used methods to determine the orbital parameters of EBs usually rely on some generative model (such as a two-body Keplerian orbit) which takes a vector of parameters which can

be called 'θ'. This generative model is incorporated into a likelihood function which computes the similarity of the model synthetic data with the observed data using a statistical metric such as the χ^2 parameter. However this process, referred to as 'Maximum Likelihood' (hereafter ML) estimation, has a number of shortcomings, such as: ML infers the mode of a distribution which is not as robust to skewed distributions as the median of the distribution. ML does not directly generate uncertainties on the inferred parameter modes. ML does not take into account any relevant prior information that may help better infer the desired parameters.

We can correct for these issues by using Bayes Theorem (Bayes 1763) to perform our inference which is written out as $P(\theta|(D) = P(D|\theta) \cdot P(\theta)/P(D)$. Here $P(D|\theta)$ is our likelihood function and $P(\theta)$ is our prior function. $P(D)$ is called the 'Bayesian Evidence' and while it is an important quantity, it is also usually very computationally expensive to calculate and primarily useful in comparing different models. We can safely ignore $P(D)$ in this work as our framework currently only uses one orbital model.

The Bayes theorem equation is fundamentally an integral over the whole of the parameter space we are interested in. Fully evaluating the integral is usually a computationally intractable endeavor. However we can evaluate this equation numerically using Markov Chain Monte Carlo (hereafter, MCMC). MCMC is an iterative way to evaluate $P(\theta|D)$ (known as the posterior distribution) by drawing samples for θ. The probabilities of the values in θ are calculated in the prior. These values are then used in our generative model and compared with the data in the likelihood function.

Most MCMC methods rely on the 'Metropolis-Hastings' algorithm (otherwise known as 'random-walk') (Metropolis et al. 1953) to draw samples to construct the posterior distributions. Unfortunately, the efficacy of MCMC relies on the presumption of sufficiently exploring the parameter space from which to infer the posterior distributions of variables with a finite number of steps. MCMC algorithms that rely on a random walk to explore a parameter space usually suffer from an incapacity to identify whether effective exploration of the target distribution is being achieved. Characterizing an MCMC run as 'converged' then typically relies on post-MCMC sampling techniques such as calculating the auto-correlation length of samples to determine if the number of samples drawn is enough; this process may need to be repeated several times to properly derive posteriors for very geometrically complicated problems.

HMC (Duane et al. 1987) methods rely on calculating the gradients within a parameter space in order to explore it efficiently (see Neal 2012). The NUTS algorithm (Hoffman & Gelman 2014) is a variation of HMC that allows researchers to bypass the usually time consuming process of performing multiple iterations of sampling with varying step size ϵ and number of steps L values to determine how best to explore a parameter space. The NUTS algorithm actively adjusts ϵ and L during sampling as well as terminating sampling when the algorithm confirms that the parameter exploration is beginning to double-back onto itself in order to maintain computational efficiency.

3.2. *Assumptions in our framework*

Jointly modeling a binary with its light curve flux measurements as well as radial velocity measurements makes it possible to infer all of the binary orbital parameters individually. We make certain assumptions to develop a clear structure to the solution for the problem. The assumptions we make with respect to the APOGEE radial velocity data are similar to the assumptions made in Price-Whelan et al. (2017), only now we use the more recent DR17 APOGEE data from Abdurro'uf et al. (2022).

Our assumptions about the radial velocity data summarily concern the time dependent gravitational pull from a single unseen companion acting on the observed star within a detached two-component binary. The assumptions on the light curve data similarly concern a completely-detached, eclipsing, two-component binary that does not have significant stellar activity contributing to the flux variability. For both the individual flux measurements and the individual radial velocity measurements we assume that the noise is Gaussian with variances set by the reported measurement uncertainties.

In detail, our assumptions about the APOGEE radial velocity data are:

1. The radial velocity data represent unbiased measurements of only the more luminous star in a binary system.

2. The time dependence of the radial velocity measurements originates only from movement of the observed star caused by the gravitational pull of a (spectroscopically-unseen) secondary star (i.e. there are no internal processes in the star which may produce a significant RV signal).

3. The noise model for the radial velocity measurements is Gaussian.

Our assumptions about the TESS light curves are:

1. The primary (secondary) eclipses in the light curve are the result of a primary star being eclipsed by a secondary star (and vice-versa) as they orbit around their common center of mass.

2. Limb-darkening effects from either star can be adequately described by a two-parameter quadratic function (Espinoza & Jordán 2015, 2016).

3. The out-of-eclipse flux of each binary's light curve can be effectively modeled by a Gaussian process employing a stochastically-damped harmonic oscillator (SHO) kernel.

4. The individual light curve flux measurements can be viewed as samples from a Gaussian distribution with variances given by the reported flux uncertainties.

3.3. *Phase parameter estimates from TESS light curves*

We must first convert all of the measurements (radial velocity and flux) to the same epoch, as well as sort the data points within each set of measurements to be chronologically in order, and zeroed out such that both sets of observation times are centered with respect to either the APOGEE epoch time or TESS epoch time. We already have two estimates of these parameters: The *The Joker* results in Price-Whelan et al. (2020) (`MAP_P` and `MAP_T0`) and the TESS BLS-derived parameters (see Section 2).

We plot all of the phased radial velocity and flux data using both the period, and epoch time (t_0) estimates from both the APOGEE MAP parameters from the *The Joker* results for that system, as well as the BLS-derived period and epoch time from the TESS light curve. We visually inspect the quality of the phase curves folded and centered by each different eclipse epoch time estimate. We find in 11/12 systems that the TESS light curve estimates for period and t_0 are more successful at folding both sets of data (TESS and APOGEE) into tightly defined phase curves.

3.4. *Setting informed priors using* Isochrones

We set the the priors of log primary mass ($\log M_1$), log primary radius ($\log R_1$), log mass ratio ($\log q$), log surface brightness ratio ($\log s$), and log radius ratio ($\log k$) using values derived from the Isochrones package (Morton 2015). Isochrones is a python package that wraps data from the MESA Isochrones & Stellar Tracks (MIST, also see Choi et al. 2016; Dotter 2016) with additional functions to perform stellar parameter fitting to single stars as well as binary stars.

We use the Isochrones 'Binary Star Model' to infer binary component stellar parameters from the available photometry of our 12 SEB systems. `BinaryStarModel` performs nested sampling using the MultiNest (Feroz et al. 2009) package which is wrapped into the PyMultiNest (Buchner et al. 2014) package. Nested sampling performed by MultiNest allows for quick, efficient sampling of the parameter space possibly multi-modal posterior distributions. The Binary Star Model fits the available photometry under the assumption that each SEB is an unresolved binary in each the pass-bands.

We query and download the available 2MASS J, H, K magnitudes (Skrutskie et al. 2006), the *Gaia* eDR3 G, G_{BP}, and G_{RP} magnitudes (Riello et al. 2020), as well as the TESS magnitude (Sullivan et al. 2015) data for each binary system in our sample. The calculated TESS magnitude is retrieved from the header data of the LightKurve data for each TIC system.

The Binary Star Model has support for custom prior distributions for distance, age, [Fe/H], mass, AV[2], and a special parameter used internally by Isochrones called the Equivalent Evolutionary Phase (EEP). The EEP is used to approximate stellar evolution along the non-domain, curvilinear main-sequence as well as the later stellar evolutionary stages of a star. We set the distance prior to be a Normal distribution centered on the photo-geometric distance (`r_med_photogeo`) derived by Bailer-Jones et al. (2020) using *Gaia* eDR3 photometry and astrometry. The spread of this distribution is set to 3× the average of the two inferred quantiles of the photo-geometric distance (`r_lo_photogeo` and `r_hi_photogeo`)

We use the available APOGEE measurements of [Fe/H], $T_{\rm eff}$, and $\log g$ to truncate the eventual parameter space that Isochrones nested sampler explores. Truncating the parameter space helps the nested sampling reach convergence faster. The posterior distributions generated from this sampling will be used as initial values for the priors we ultimately use in our PyMC3 framework to model SEBs.

We perform a rudimentary grid search with Isochrones to set informed limits on the sampling priors Isochrones uses in the Binary Star Model. We use the Isochrones `MISTEvolutionTrackGrid` interpolator to bootstrap sample across a grid of isochrones spanning the Isochrones EEP range after the pre-main sequence phase [202-1700] and masses spanning [0.1 $M_\odot$–50 $M_\odot$, $M_\Delta = 0.025$ $M_\odot$]. We set the metallicity [Fe/H] value to the APOGEE measured [Fe/H]. The $T_{\rm eff}$ and $\log g$ APOGEE measurements are then used to perform acceptance/rejection of the grid search generated isochrones. We keep the generated isochrones which have a $T_{\rm eff}$ and $\log g$ value within the APOGEE measured $T_{\rm eff} \pm 500$ K and $\log g \pm 0.5$ dex, respectively. The bounds for the uniform priors Isochrones uses for mass, age, and EEP are set to the min/max ranges calculated from the accepted isochrones.

We run the Isochrones nested sampler with the aforementioned bounds and run until convergence. We take the primary component mass ($\log M_1$) to be the greater of the two masses derived with Isochrones. We calculate the surface brightness of each component according to Graczyk et al. (2017, Equations 2 & 3) to get the surface brightness ratio, $\log s$. We construct $\log q$ and $\log k$ as the ratio of

[2] The total extinction in the V band within the Johnson-Morgan UBV system

Table 1. Priors used in the joint RV–LC orbit model (this work)

Parameter Name	Distribution	Hyper-parameters	Description
$\sigma_{\log m_1}, \sigma_{\log r_1}$	Inverse Gamma	(7, 0.25)	SD of M_1, R_1 priors
$\sigma_{\log k}, \sigma_{\log q}, \sigma_{\log s}$	Inverse Gamma	(6, 0.5)	SD of $\log k$, $\log q$, $\log s$ priors
$\log M_1$	Normal	$(\mu_{\log m_1}, \sigma_{\log m_1})$	Mass of primary
$\log R_1$	Normal	$(\mu_{\log r_1}, \sigma_{\log r_1})$	Radius of primary
$\log q$	Normal	$(\mu_{\log q}, \sigma_{\log q})$	Ratio of component masses
$\log s$	Normal	$(\mu_{\log s}, \sigma_{\log s})$	Ratio of component surface brightness
$\log k$	Normal	$(\mu_{\log k}, \sigma_{\log k})$	Ratio of binary component radii
b	Uniform	$[0, 1+k]$	Impact parameter
(e, ω^a)	Unit-Disk	$(\sqrt{e}\cos\omega_{est}, 0.0)$	Unit disk function
t_0	Normal	(t_0, 3 d)	Time of the first eclipse
t_N	Normal	(t_N, 3 d)	Time of the Nth eclipse
u_1	–	$[u_{1,0}, u_{1,1}]^b$	Limb darkening coefficients of primary
u_2	–	$[u_{2,0}, u_{2,1}]^b$	Limb darkening coefficients of secondary
$\mu_{\rm RV}$	Normal	(0, 50 km s^{-1})	Mean background radial velocity
$\mu_{\rm LC}$	Normal	(0, 10 ppt)	Mean background light curve flux
$\sigma_{\rm GP}$	Inverse Gamma	†	SD of Gaussian process
$\rho_{\rm GP}$	Inverse Gamma	†	Un-damped period of Gaussian process
$\sigma_{\rm LC}$	Inverse Gamma	†	Light curve flux 'jitter' term
$\log \sigma_{\rm RV}$	Truncated Normal	($\log(\widetilde{\sigma_{\rm RV}})$, 10 km s^{-1})	Radial velocity jitter term

[a] - argument of periastron

[b] - Uninformative prior for quadratic limb-darkening parameters as implemented by Kipping (2013)

† - We utilize the inverse gamma distribution parameter estimation algorithms outlined by Michael Betancourt https://betanalpha.github.io/assets/case_studies/gp_part3/part3.html#4_adding_an_informative_prior_for_the_length_scale

the secondary component value over the primary component value for mass and radius, respectively. We sigma clip the $\log M_1$, $\log R_1$, $\log q$, $\log k$, and $\log s$ distributions to within 5 σ. We use the median absolute deviation (MAD) of each distribution multiplied by a factor of 1.4286 to perform sigma clipping. The MAD parameter is more robust to outliers than the standard deviation. The scale parameter of 1.4286 is used so the MAD more closely approximates the standard deviation the of normal distribution (see Leys et al. 2013). The μ, σ values from the sigma clipped $\log M_1$, $\log R_1$, $\log q$, $\log k$, and $\log s$ distributions are used to generate the values for the priors we use in our SEB orbital model (see Section 3.5).

3.5. *Constructing the Spectroscopic Eclipsing Binary Orbital Model*

Our priors will be used to construct the orbital model within the Exoplanet python module (Foreman-Mackey et al. 2021), along with relevant noise terms. We compute the likelihood of the parameters of this orbital model using both the APOGEE-measured radial velocities and the TESS

light curve flux measurements. The $\log M_1$, $\log R_1$, $\log q$, $\log k$, and $\log s$ priors have μ values that are set using the μ values derived from Isochrones (see Section 3.4). We let the variance of these five priors vary within the Exoplanet model by setting all five variances with two inverse Gamma distribution priors. We set one inverse gamma prior for the $\log M_1$ and $\log R_1$ variances, and another inverse gamma prior for the three SEB ratio ($\log k$, $\log q$, and $\log s$) variances. The inverse gamma priors are initialized with σ values from Isochrones priors multiplied by a large factor of 256, which we set arbitrarily, to account for possible biases within the Isochrones framework arising from its use of the MIST stellar model grid fitting under an assumption of blended photometry.

We employ a Gaussian process using the Celerite2 package (Foreman-Mackey et al. 2017; Foreman-Mackey 2018) to model the out-of-eclipse light curve flux of each eclipsing system. We use the SHO kernel in our Gaussian process. The two parameters that dictate the behavior of our Gaussian process are $\rho_{\rm GP}$, the un-damped period of the SHO, and $\sigma_{\rm GP}$, the standard deviation of the process. We also include a $\sigma_{\rm LC}$ prior to model the jitter present in the TESS light curve flux measurements. We only have one noise parameter for the radial velocity measurements, $\log \sigma_{\rm RV}$, to model the jitter present in the APOGEE radial velocity measurements. We set $\log \sigma_{\rm RV}$ with a log-normal prior, bounded at an upper value of $10 {\rm km\,s^{-1}}$.

Figure 2. A visualization of the light curve of TIC 20215452 to demonstrate the usage of a Gaussian process to remove the background flux from a light curve. **Top**: A small section of the unfolded light curve of this system shown as colored markers, with the unfolded Gaussian process plotted as the dashed (gold) line, and the unfolded eclipse model over-plotted as the solid (cyan) line. **Middle**: The complete folded phase curve (markers, colored by absolute time), the folded Gaussian process model (group of dotted lines), and the folded eclipse model (colored solid line). The Gaussian process and the light curve data points are colored by absolute time in all panels. **Bottom**: The folded light curve (colored markers) with the fitted Gaussian process subtracted from the TESS data points. The light curve model is over plotted in cyan. Using a Gaussian process to model the background light curve flux allows for generalized fitting of multiple types of stellar variability. **Fig. Set 2. TESS light curve and GP decomposition plots**

We employ a unit disk function built into Exoplanet to sample ($e\cos\omega$, $e\sin\omega$). We then calculate the eccentricity e, and argument of periastron ω as $(e\cos\omega)^2 + (e\sin\omega)^2$, and $\arctan\frac{e\sin\omega}{e\cos\omega}$, respectively. We provide an estimate of $e\cos\omega$ to our model by performing a two stage BLS fit to the TESS light curve. We use the built-in eclipse masking capabilities of the LightKurve module to mask out the primary eclipse of each binary's TESS data and refit a second BLS to estimate eclipse parameters for the secondary eclipse.

We derive an estimate for $e\cos\omega$ by rearranging Equation 33 from Winn (2010) using the eclipse epoch time $T_{0,1}$ for the primary eclipse and the eclipse epoch time $T_{0,2}$ for the secondary eclipse. The $e\cos\omega$ value we derive and $e\sin\omega$, which we set to 0.0, are used to initialize the Exoplanet unit-disk function from which we derive values of eccentricity e and argument of periastron ω.

The impact parameter b is modeled internally using the Exoplanet impact parameter distribution function, which is conditioned on the radius ratio k (not to be confused with the radial velocity semi-amplitude K). We use quadratic limb darkening parameters (Kipping 2013) for both the primary and the secondary light curve eclipses in our model.

We parameterize the orbital period P of the binary as a function of the epoch time of first eclipse t_0, and the last epoch of eclipse t_N, such that $P = \frac{(t_N - t_0)}{N_{\rm eclipses}}$. This parameterization of the period removes

a degeneracy present between period and t_0. This degeneracy arises from the long time baselines in the TESS observations and the shortness of the period of these binaries ($\lesssim 50$ days) which introduces a numerical sensitivity in the calculation of t_N. We model both the mean light curve flux and mean radial velocity present in each set of observations with normal distribution priors with a μ of zero and a σ of 10 ppt and 50 $\mathrm{km\,s^{-1}}$, respectively.

3.6. *Calculating the Maximum a posteriori values to initiate sampling*

We use PyMC3 to calculate the Maximum *a posteriori* (hereafter, MAP) estimates of our orbital parameters. The MAP estimates are useful to initiate sampling within a parameter space which bolsters the HMC sampler's ability to construct reliable posterior distributions moreso than random initializations.

We perform MAP estimation steps on individual parameters in a sequential order before estimating all 19 parameters simultaneously. Sequential MAP estimates increase the numerical stability of the final MAP estimate of all 19 parameters. The order of the sequence is allowed to vary between different SEB systems to increase the flexibility of the framework.

The first MAP estimations that take place occur on radius ratio '$\log k$', which greatly influences the depth of the two eclipse dips. The impact parameter 'b' is next optimized as it influences the eclipse duration in our orbital model. The unit disk function 'ecs' is then optimized as the eccentricity calculated from it potentially influences the time between the primary and secondary eclipse as well as also affect the shape of the model fit to the radial velocity data.

The MAP parameter estimates for [(t_0, t_N), (u_1, u_2), $\log M_1$, $\log R_1$, $\log s$, $\log q$, mean-rv, mean-lc] are ordered to maximize the gradient of the likelihood. This order is calculated in a brute force manner, looping through the set of parameters until the MAP step resulting in the maximum positive change in the log-likelihood is found. The parameter considered in this MAP step is then removed from the set and the loop is restarted. This loop continues until all parameters in the set are optimized.

The MAP estimates of the σ_{GP}, σ_{LC}, and ρ_{GP} parameters are performed simultaneously in a single step. This allows the eclipse model has been fit to the data as much as possible before attempting to fit the Gaussian process to the out-of-eclipse background flux. Performing these MAP estimates before the eclipse model is fully fit to the light curve data may lead to the Gaussian process over-fitting. The $\log \sigma_{\mathrm{RV}}$ parameter is also optimized after the completion of the brute force loop to avoid over-estimating the jitter in the radial velocity background.

The two inverse gamma priors which set the variances for $\log M_1$, $\log R_1$, $\log k$, $\log q$, and $\log s$ are not optimized until the final MAP step. The very broad initial variance parameters on the five orbital parameter priors remain throughout the previous steps to maximize flexibility in estimating the MAP values for all parameters. All 19 orbital model parameters have their MAP estimations performed simultaneously in the final step.

We iteratively perform sigma clipping to remove outliers from the SEB light curve. We calculate the residuals of the light curve model with the TESS data and mask light curve data points with $\|residuals\| > 5\sigma$. MAP values are recomputed after each sigma clipping of the light curve for a maximum of 10 iterations or until no light curve data points are masked.

3.7. *Sub-sampling the light curve using the eclipse model*

We use the MAP values computed after sigma clipping is completed to sub sample the out-of-eclipse flux to reduce computational time of sampling. The out-of-eclipse flux in a light curve is not simply background. It contains important information about the stars such as stellar variability, surface pulsations, and rotation signatures. Our framework models the background flux with a Gaussian process and treats this flux variability as "nuisance" or additional, correlated noise. We acknowledge that there is information to be distilled from the background flux and may integrate that functionality into our framework at a later time.

We perform targeted sub-sampling of the light curve background flux for our sample of SEB systems. This step is performed to (a) reduce computational cost of evaluating the likelihood across the entire light curve observation for a binary system with every sampling step, and (b) reduce density of background flux to increase eclipse visibility, as in the case for TIC 144441148 (Figure 1, row 2, column 2).

We mask out the primary and secondary eclipses along the entire light curve using the eclipse model with the MAP values estimated in the previous sub section. We sub-sample the out-of-eclipse light curve flux data with a random uniform draw down to a specified size. We find that sub-sampling the light curves to ∼5000 data points provides the best balance of computational efficiency and sampling reliability for all 12 SEB systems.

3.8. *Generating posterior samples with the Exoplanet orbital model*

We perform sampling using the priors and likelihood function in the Exoplanet model with 6 sampling chains. Each chain is set to run for 5000 tuning steps and 1000 draw steps. We consider each model to have converged upon the completion of sampling when the rank normalized split-$\hat{r}$ values for all of the model sampling parameters are below 1.01 (see Vehtari et al. 2019). The rank normalized split-$\hat{r}$ value improves upon the split-$\hat{r}$ () diagnostic's ability to detect convergence between multiple sampling chains.

4. RESULTS

Our framework successfully infers complete orbital/stellar parameter posterior distributions for 11 of our SEB systems and a partial orbital/stellar parameter posterior distributions for 1 SEB system[3]. We plot both the radial velocity phase curves and the GP-subtracted, eclipse-phased light curves for all 12 fitted systems using the median orbital model parameters in Figure 3. We also plot the 68th and 95th confidence intervals of the orbital model in both the phase curves in Figure 3. We plot an orrery of the 12 SEBs with a top-down view of their orbits generated from the median of parameter posteriors in Figure 4 to visualize the different binary component configurations modeled with our framework.

We also list a summary of important binary parameters in Table 2. The super-scripts (sub-scripts) listed with each column in 2 are the upper (lower) bound uncertainty calculated from the Bayesian Credible Interval (hereafter, BCI) for each parameter. The BCI for a parameter can be interpreted as the interval of the marginal posterior PDF within which the true parameter value falls with a specified probability (see Dahn 1978). The BCI probability is set to 0.68.

The median mass of the primary components in our SEB sample is $1.06 M_{\odot}$ with a 16th and 84th percentile of $0.75 M_{\odot}$ and $1.31 M_{\odot}$, respectively. The median mass of the secondary components is

[3] TIC 144441148, unreliable secondary component radius inference

Table 2. Relevant posterior distribution values for orbital parameters of the 12 SEBs considered in this work, ordered in ascending TIC number. The uncertainties for all parameters listed are calculated from the Highest Density Interval (HDI) for each parameter posterior distribution at a 68% probability (The HDI can be considered as the minimum width Bayesian Credible Interval).

TIC-ID	M_1	R_1	M_2	R_2	P	a	e	q
–	$M_\odot$	$M_\odot$	$R_\odot$	$R_\odot$	d	$R_\odot$	–	–
20215452	$0.69^{+0.04}_{-0.04}$	$0.79^{+0.02}_{-0.02}$	$0.33^{+0.02}_{-0.01}$	$0.35^{+0.01}_{-0.01}$	$0.59576^{+8e-07}_{-7e-07}$	$3.0^{+0.05}_{-0.04}$	$0.0112^{+0.0063}_{-0.0112}$	$0.48^{+0.01}_{-0.02}$
28159019	$1.23^{+0.05}_{-0.05}$	$1.99^{+0.04}_{-0.04}$	$0.34^{+0.01}_{-0.01}$	$0.35^{+0.01}_{-0.01}$	$4.61877^{+3.6e-06}_{-4.1e-06}$	$13.57^{+0.18}_{-0.17}$	$0.0018^{+0.0009}_{-0.0018}$	$0.28^{+0.0}_{-0.01}$
99254945	$0.77^{+0.03}_{-0.03}$	$0.71^{+0.01}_{-0.01}$	$0.33^{+0.01}_{-0.01}$	$0.32^{+0.0}_{-0.0}$	$7.70784^{+2.83e-05}_{-2.58e-05}$	$16.97^{+0.19}_{-0.19}$	$0.0052^{+0.0005}_{-0.0004}$	$0.43^{+0.01}_{-0.01}$
126232983	$0.89^{+0.03}_{-0.03}$	$1.65^{+0.09}_{-0.08}$	$0.14^{+0.0}_{-0.0}$	$0.14^{+0.01}_{-0.01}$	$4.53432^{+3.91e-05}_{-4.35e-05}$	$11.66^{+0.14}_{-0.14}$	$0.003^{+0.0014}_{-0.003}$	$0.16^{+0.0}_{-0.0}$
144441148	$0.82^{+0.03}_{-0.04}$	$1.76^{+0.12}_{-0.09}$	$0.14^{+0.0}_{-0.0}$	$0.71^{+0.08}_{-0.08}$	$5.81856^{+5.07e-05}_{-4.03e-05}$	$13.41^{+0.14}_{-0.18}$	$0.0169^{+0.0037}_{-0.0031}$	$0.17^{+0.0}_{-0.0}$
164458426	$1.54^{+0.06}_{-0.06}$	$1.93^{+0.05}_{-0.09}$	$0.2^{+0.01}_{-0.0}$	$0.21^{+0.01}_{-0.01}$	$4.69331^{+1.35e-05}_{-1.29e-05}$	$14.2^{+0.19}_{-0.15}$	$0.0022^{+0.0011}_{-0.0021}$	$0.13^{+0.0}_{-0.0}$
164527723	$1.15^{+0.04}_{-0.04}$	$1.85^{+0.05}_{-0.04}$	$0.18^{+0.01}_{-0.01}$	$0.26^{+0.01}_{-0.01}$	$2.73987^{+3e-06}_{-2.9e-06}$	$9.06^{+0.12}_{-0.1}$	$0.0097^{+0.0109}_{-0.0097}$	$0.16^{+0.01}_{-0.01}$
165453878	$1.07^{+0.05}_{-0.04}$	$1.14^{+0.02}_{-0.02}$	$0.2^{+0.01}_{-0.01}$	$0.22^{+0.0}_{-0.0}$	$2.7076^{+5.3e-06}_{-5.3e-06}$	$8.84^{+0.14}_{-0.11}$	$0.0019^{+0.0017}_{-0.0019}$	$0.19^{+0.0}_{-0.0}$
169820068	$1.2^{+0.05}_{-0.05}$	$4.29^{+0.08}_{-0.09}$	$1.04^{+0.02}_{-0.02}$	$1.13^{+0.03}_{-0.03}$	$29.74447^{+0.0001755}_{-0.0001856}$	$52.85^{+0.58}_{-0.52}$	$0.01^{+0.0012}_{-0.0011}$	$0.87^{+0.01}_{-0.02}$
258108067	$0.69^{+0.03}_{-0.02}$	$1.62^{+0.02}_{-0.02}$	$0.44^{+0.01}_{-0.01}$	$0.5^{+0.01}_{-0.01}$	$12.96863^{+0.0001022}_{-0.0001597}$	$24.14^{+0.24}_{-0.24}$	$0.3477^{+0.001}_{-0.0013}$	$0.63^{+0.01}_{-0.01}$
271548206	$1.05^{+0.04}_{-0.04}$	$0.93^{+0.02}_{-0.02}$	$0.36^{+0.01}_{-0.01}$	$0.42^{+0.01}_{-0.01}$	$7.44835^{+1.41e-05}_{-1.21e-05}$	$17.99^{+0.19}_{-0.21}$	$0.0322^{+0.0022}_{-0.0019}$	$0.35^{+0.01}_{-0.0}$
272074664	$1.61^{+0.1}_{-0.07}$	$1.9^{+0.04}_{-0.03}$	$0.7^{+0.02}_{-0.02}$	$0.63^{+0.01}_{-0.01}$	$10.47626^{+5.7e-06}_{-5.8e-06}$	$26.63^{+0.45}_{-0.34}$	$0.0546^{+0.0005}_{-0.0005}$	$0.44^{+0.01}_{-0.01}$

NOTE—The secondary component radius of TIC 144441148 has been colored red to indicate an unreliable inference by the model on the secondary radius.

$0.32M_\odot$ with a 16th and 84th percentile of $0.17M_\odot$ and $0.49M_\odot$, respectively. There are 10 SEB systems which have a median eccentricity $\lesssim 0.05$ but BCIs largely consistent with zero. SEB system (TIC 285108067) is the only non-circular orbit in our sample, with a median eccentricity of 0.35.

Our framework infers tight posterior distributions for the component masses in each SEB. The primary masses in our sample of 12 SEBs have a median uncertainty of 3.81%, with a 16th and 84th percentile of 3.68% and 4.63%, respectively. The secondary masses have a median uncertainty of 2.54%, with a 16th and 84th percentile of 2.20% and 3.80%, respectively.

The median values of the orbital solution period largely agree with the orbital periods derived from the light curve BLS periodigram analysis. The median orbital period of our SEB sample is 5.25 days with a 16th and 84th percentile of 2.73 days and 11.07 days, respectively. We briefly describe the inferred parameters and orbital structure of the 12 systems in our sample below, ordered by their `TIC` ID.

TIC 20215452: This system has the shortest orbital period of the systems in our sample, with a period of about $P \approx 0.60$ days. The median of the eccentricity posterior distribution is about

0.01. The computed semi-major axis of the system is 2.99 $R_\odot$. The primary component has a mass of $M_1 \approx 0.69\ M_\odot$. The mass ratio q[4] of this system is about $q \approx 0.48$.

TIC 28159019: This system has a significantly curved primary eclipse, while its secondary eclipse displays a flatter eclipse. This system has the lowest inferred eccentricity in our sample with a posterior distribution mode of 0.002 and a credible interval on the eccentricity posterior that is consistent with zero. The computed semi major-axis of the system is 13.57 $R_\odot$. The orbital period of this system is $P \approx 4.62$ days. The primary component mass is $M_1 \approx 1.23 M_\odot$ and the system's mass ratio is $q \approx 0.28$.

TIC 99254945: This binary system has a computed orbital period of about $P \approx 7.71$ days. The semi-major axis of this system has been calculated to be 16.97 $R_\odot$. The median of the eccentricity posterior distribution is found to be 0.005. The primary mass is found to be $M_1 \approx 0.77\ M_\odot$. The mass ratio for this binary system is $q \approx 0.43$. This system has the deepest observed primary eclipse (-18.52%) in our sample.

TIC 126232983: This binary system has a very shallow and inconspicuous secondary eclipse. The orbital period of this system is about $P \approx 4.53$ days. The median of the eccentricity posterior distribution is about 0.003 but the credible interval is consistent with zero. The semi-major axis of this system has been computed to be 11.66 $R_\odot$. The primary component mass is $M_1 \approx 0.89\ M_\odot$. The mass ratio is $q \approx 0.161$.

TIC 144441148: This binary system is the only binary system in our sample which has a non-existent secondary eclipse in the light curve. The orbital period for this system was computed to be $P \approx 5.82$ days. The semi-major axis was computed to be 13.41 $R_\odot$. The median of the eccentricity posterior distribution was found to be 0.02. The primary component mass was found to be $M_1 \approx 0.82\ M_\odot$. The mass ratio is $q \approx 0.17$. Our framework infers a secondary component mass of $M_1 \approx 0.14\ M_\odot$ but a radius of $R_2 \approx 0.71\ M_\odot$. We take this radius inference as un-reliable as it is non-physical, as seen on the mass-radius plot in Figure 5 (black triangle). We discuss this un-reliable inference further in Section 5.

TIC 164458426: This binary system has some interesting periodic signals in the background flux that prohibit a visual observation of the secondary eclipse without subtraction of the background flux by a Gaussian process. The orbital period was computed to be $P \approx 4.69$ days. The median of the eccentricity posterior distribution was found to be 0.002 but the credible interval is consistent with zero. The semi-major axis was computed to be 13.42 $R_\odot$. The primary component mass was found to be $M_1 \approx 1.54\ M_\odot$. The mass ratio is $q \approx 0.13$.

TIC 164527723: This binary system has a computed orbital period of $P \approx 2.71$ days. The semi-major axis was computed to be 9.06 $R_\odot$. The median of the eccentricity posterior distribution was found to be 0.010 but the credible interval is consistent with zero. The primary component mass was found to be $M_1 \approx 1.10\ M_\odot$ and the mass ratio is $q \approx 0.16$.

TIC 165453878: This binary system has a computed orbital period of $P \approx 2.71$ days. The median of the eccentricity posterior distribution is found to be 0.002 but with a credible interval

[4] We define q to be (M_2/M_1)

Figure 3. Plotted phased radial velocity - light curve for the 12 modeled TESS–APOGEE SEBs in this work. Even rows are the radial velocity phase curves for the 12 systems. Odd rows are the eclipse light curves for the 12 systems. The data points in each system are plotted as black dots. The final orbital model in both the light curve and radial velocity phase curves is plotted in orange. The 68% and 95% confidence intervals plotted as transparent orange fill. Our framework is able to reliably infer orbital parameter posterior distributions for SEBs with varied data.

consistent with zero. The semi-major axis was computed to be $8.84 R_\odot$. The primary mass component was found to be $M_1 \approx 1.07\ M_\odot$ and the mass ratio was found to be $q \approx 0.19$.

TIC 169820068: This binary is the only system in our sample where the APOGEE-derived period and epoch eclipse time served to fold the RV phase curve and flux time-series phase curve better than the TESS light curve derived values. The orbital period was found to be $P \approx 29.74$ days, making it the longest period binary in our sample. The semi-major axis is computed to be about 52.85 $R_\odot$. The median of the eccentricity posterior distribution was found to be 0.01. The primary component mass was found to be $M_1 \approx 1.20\ M_\odot$. The secondary component mass was found to be $M_2 \approx 1.04\ M_\odot$. The mass ratio was found to be $q \approx 0.87$ This is also the only system in our sample for which the final orbital model produces a larger secondary eclipse than the primary eclipse. The primary component radius is $R_1 \approx 4.29\ R_\odot$, which is much larger than main-sequence mass-radius relations. These peculiarities raise the implication that the primary component star is an evolved star.

TIC 258108067: This binary system has the highest median eccentricity in our sample at 0.35. This system necessitated the relaxation of automatically setting a ceiling on $e \cos\omega$ of 0.01 in the initialization of the orbital model. The orbital period was computed to be $P \approx 12.97$ days. The semi-major axis is computed to be 24.14 $R_\odot$. The primary component mass and mass ratio were found to be $M_1 \approx 0.69\ M_\odot$ and $q \approx 0.63$, respectively.

TIC 271548206: This binary system has very broad background flux in the light curve data, making the secondary eclipse more than difficult to visual identify. The orbital period was computed to be $P \approx 7.45$ days. The semi-major axis was computed to be 17.99 $R_\odot$. The median of the eccentricity posterior distribution was found to be 0.03. The primary component mass was found to be $M_1 \approx 1.05\ M_\odot$. The mass ratio was found to be $q \approx 0.35$.

TIC 272074664: This binary system has a computed orbital period of $P \approx 10.48$ days. The semi major was computed to be 26.63 $R_\odot$. The median of the eccentricity posterior distribution was found to be 0.05. The primary component mass was found to be $M_1 \approx 1.60\ M_\odot$, which is the largest in our sample. The mass ratio was found to be $q \approx 0.44$.

5. DISCUSSION

We demonstrated here the flexibility of our framework to derive complete orbital parameter posterior distributions for a sample of 12 different SEBs with a varying number of radial velocity measurements and different light curve eclipse geometries. There are 11 binary systems in our sample did not have model framework settings that required further manual fine-tuning before sampling took place. Only one system, TIC 285108067, required manual intervention in relaxing the circular eccentricity

Figure 4. Face-on plot of all 12 TIC SEBs considered in this work, ordered by increasing TIC-ID. The primary component is plotted on the blue orbit, the secondary component is plotted on the orange orbit. The relative symbol sizes of all binary components are to scale with each other. The relative sizes of plotted component symbols are not to scale to the plotted orbits. Each plot has a small scale indicator of either 1 AU, 5 AU, or 10 AU. Each SEB has it's TIC-ID listed below it's orbit.

setting. Our framework only requires a TESS TIC ID in order to be initialized, allowing a list of EB systems to be modeled simultaneously.

The final orbital models constructed from our inferred orbital parameters and plotted over the TESS + APOGEE data in Figure 3 highlight the varied configurations of SEBs that our framework is able to model. Figure 3 also illustrates the power of jointly considering the TESS + APOGEE data. In the case of a very noisy out-of-eclipse flux and a very shallow secondary eclipse, such as TIC 144441148, the well sampled radial velocity measurements likely bolsters the framework's ability to constrain the orbital parameters.

Our framework is also capable of modeling SEBs with sparsely observed secondary eclipses which is the case for the systems TIC 169820068 and TIC 258108067 (see Figure 8 in the Appendix). TIC 258108057, in particular, has a number of characteristics that highlight the flexibility of our framework: it has only 5 radial velocity measurements, and has a sparsely observed secondary eclipse in the light curve data, and is an eccentric binary.

We have plotted the mass-radius contours created from the posterior distributions of the 12 SEBs we modeled in this work in Figure 5. We have plotted the masses and radii of the 95 eclipsing binaries which were characterized in Torres et al. (2010b) in Figure 5 as well. The Torres et al. (2010b) binaries trace out a mass-radius main sequence that most of the median mass and radii estimates for our sample fall within as well.

We have marked out the minimum mass and radius measured in the Torres et al. (2010b) EB sample in Figure 5 as gray dash lines. The inferred primary masses and radii in our sample fall within this range of masses and radii. There are 4 secondary components which have masses and radii that fall below these ranges. These lower masses and radii highlight the higher overall sensitivity that is achievable by jointly modeling both the TESS + APOGEE data with a Bayesian probabilistic framework.

5.1. *Spectroscopic Eclipsing Binaries populations as benchmarks*

Recently there has been increased work into measuring the parameter distributions of binary populations using Bayesian hierarchical inference. Bayesian hierarchical inference is performed similarly to 'standard' Bayesian inference with a prior and a likelihood informing a posterior. The 'hierarchical' aspect refers to the prior being modifiable during sampling with a 'hyper-prior' informing on what shape the prior distribution might have. The prior distribution shape is then inferred as sampling takes place.

Bayesian hierarchical inference has been used to explore the a number of different characteristic distributions important to the understanding of binaries. Price-Whelan et al. (2020) measured the long period binaries eccentricity distributions with results from *The Joker*. Lyttle et al. (2021) measured the distributions of stellar helium abundance (Y) using Kepler dwarfs and sub-giants in

a Bayesian hierarchical model. Hwang et al. (2022) used Bayesian hierarchical inference to measure the eccentricity distribution of wide binaries using astrometric data from *Gaia* DR3.

Our framework generates full posterior distributions for all binary orbital parameters rather than just a single value for the peak (or median) and the range of an appropriate interval. The complete posterior distributions of these orbital parameters can be used to perform more robust Bayesian hierarchical inference. The shape of the posterior distributions of each population member in a Bayesian hierarchical inference can better inform the shape of the hyper-prior of interest.

Our framework is able to infer full orbital parameter estimates for SEBs with single-component radial velocities. Single-lined SEBs are the more commonly observed than double-lined SEBs. This arises from the lower probability of observing shifts from both stars in the spectra unless directed long-term spectroscopic observations are carried out. Double-lined SEBs are also not as frequently observed as single-lined SEBs since single-lined SEBs occur more frequently across the range of mass ratio 'q'. Double-line SEBs are usually only observed when the masses of both components are near-equal. The more frequently occurring, and more easily observed single-lined SEBs can be used as population level calibrators for stellar evolution models.

Our framework currently only focuses on modeling single-lined SEBs, and in future work we will expand our framework to also model double-lined SEBs. The addition of a secondary set of radial velocity measurements will result in tighter posterior distributions of the inferred parameters. This will only increase our framework's capabilities for flexible SEB parameter inference and its potential for binary population parameter inference.

5.2. *Comparison with results from Gaia DR3*

The recent release of the *Gaia* DR3 (Prusti et al. 2016; Vallenari et al. 2022) there has been a massive quantity of eclipsing, astrometric, and spectroscopic binary data released taken by the *Gaia* spacecraft and processed by the *Gaia* collaboration. We compare some of our orbital parameters and stellar parameters for the binary systems which were bright enough in the *Gaia* passbands to have available data in Figure 6. In the left most panel of Figure 6 we have plotted the derived orbital period for our binary systems on the x-axis and the *Gaia* DR3 periods on the y-axis. The residuals of the two period measurements lie in the range between $\pm 2 \times 10^{-3}$ days, indicating excellent agreement between the two.

In the middle panel of Figure 6 we have plotted the orbital eccentricity for our binary systems (x-axis) as well as the *Gaia* DR3 eccentricities (y-axis). It appears that our framework infers more circularized eccentricities than the *Gaia* DR3 measurements. This is true for all the plotted binary systems except TIC 281508067 which has the highest eccentricity in our sample; both the *Gaia* DR3 eccentricity as well as our framework inferred eccentricity agree very well. Error-bars are plotted, though smaller than plot symbols for our measurements of eccentricity.

In the right most panel of Figure 6 we have plotted the inferred surface gravity, $\log g$, calculated from our framework (x-axis) for the primary stars in our sample against their $\log g$ (column=`LOGG`) measurements from APOGEE (Majewski et al. 2017) DR17 (y-axis). The spectrographic $\log g$ APOGEE measurements show good agreement within errors with the Bayesian framework derived $\log g$ values derived in this work.

5.3. *Possible radius inflation in secondary components*

There is a developing consensus that there are observed deviations in measured radii of low-mass stars ($<1\ M_\odot$) from the radii predicted by stellar evolutionary models. The mechanism that generates this observed 'radius inflation' is still not entirely clear but it's believed to be originating from strong magnetic fields interrupting the convective process, large star spots affecting the pressure and temperature of a star, or a combination of both. Radius inflation can cause a single star to move vertically on the Hertzsprung-Russell diagram because of changes in luminosity and be mistakenly identified as belonging to the binary main-sequence, thereby biasing photometric studies of binaries, especially in open clusters. Moreover, radius inflation can directly impact exoplanet characterizations as precision of the eclipse parameters from a light curve are greatly driven by the precision of the stellar radius.

There are some secondary components near the lower end of the mass range in Figure 5 that lie above the expected main sequence that is outlined by the MIST isochrones also plotted (red lines). We speculate that these secondary stars might be undergoing radius inflation and perform a rudimentary analysis to explore that possibility.

We generate several thousand theoretical radii for secondaries with component masses $<1\ M_\odot$ using the MIST isochrones in the Isochrones package. These isochrones span a uniform grid in age, consider only the main-sequence phases of stellar evolution ($202 > \mathrm{EEP} > 654$), and use the APOGEE derived [Fe/H] for each system. We calculate the median percent deviation of our framework's inferred radii from the theoretical MIST radii we generate.

We find five secondary components in our sample may be undergoing radius inflation. The secondary components in TIC 20215452 and TIC 28159019 have radii that are inflated on the order of ∼7% and ∼6%, respectively. The other three secondary components of TIC 285108067, TIC 271548206, and TIC 164527723 have larger inflated radii on the order of ∼20%, ∼16%, and ∼17%, respectively. This analysis is largely speculative and non-exhaustive but it does illustrate the versatility of this framework in inferring component radii at the low-mass end.

5.4. *Non-physical stellar parameter inferences*

Our framework is able to infer orbital parameter solutions for detached SEB systems that are largely agnostic of stellar evolutionary models. However this also opens up the framework to possibly inferring non-physical parameters in some solutions. This is the case for SEB system TIC 144441148. The median inferred mass and radius for the secondary component are $M_2 \approx 0.14\ M_\odot$ and $R_2 \approx 0.71\ R_\odot$, respectively. These stellar parameters are non-physical for a star, given its position (black triangle) on the mass-radius plot in Figure 5. The mass-radius plot can be considered an analog for the Hertzsprung-Russel diagram, and as such, there should not be stars in that area of the mass-radius space.

We consider the secondary component mass to be reliable since the component masses are derived from orbital dynamics constrained by the radial velocity measurements. TIC 144441148 is one of the SEB systems in our sample with over 20 APOGEE radial velocity measurements. The model radial velocity phase curve agrees very well with the radial velocity measurements in Figure 3[5].

The source of this non-physical solution is likely a result of the light curve data for this system, which constrains the primary component radius and secondary component radius via the radius ratio prior $\log k$. The light curve for this SEB does not appear to have secondary eclipse present, which

[5] column 1, row 3

might indicate a grazing eclipsing binary. Our framework infers a large impact parameter of $b \approx 1.38$ which further suggests a grazing eclipse for near circular orbits.

There are a number of different sources for an eclipsing binary to have a grazing orbit. A third orbiting body may induce perturbations in the orbit of the binary, affecting the eclipse depths. Stellar pulsations may cause changes in the stellar radii and affect the inference of $\log k$. Stellar pulsations can be found by considering the out-of-eclipse flux in a light curve. Our framework only considers the out-of-eclipse background flux as a nuisance parameter in its current implementation by design. Future implementations of this framework would have to address the particularities of grazing eclipsing binaries.

5.5. *Covariances within the posteriors*

We find most of the posterior distributions of the inferred orbital parameters for these systems to be well-behaved. However we do visually identify a covariance between $\log M_1$, $\log q$, and $\log R_1$. This covariance appears to be present in all posterior distributions, even in systems where the posterior distributions are well behaved (i.e. a defined unimodal shape with little to no covariance between other parameters).

We plot an example of 'well behaved' posterior distributions in the form of a corner plot for TIC 272074664 in Figure 7. In the corner plot of $\log R_1$ and $\log M_1$ (Figure 7, row 2, column 1) we can see the skewed covariance present between the these two orbital parameters.

This skewed contour shape is present after transforming to the $\log M_1$ and $\log R_1$ posterior values to M_p and R_p, respectively. Figure 5 shows the still skewed shape of the transformed $\log M_1$ and $\log R_1$ posteriors. This skewed shape is indicated that the $\log M_1$ and $\log R_1$ priors are influenced by their relation in defining the primary star's density, which is an important value attainable through light curve analysis. The shape of the secondary components mass-radius contours are also skewed despite no sampling taking place over any secondary component parameters.

The skewed shape of the secondary component's mass and radius stems from being calculated from multiplying (or adding in log space where we sample) the mass ratio q by the primary component mass. The skewed shape of the sampled $\log q$ posteriors broadcast into the calculated secondary component mass. Additionally, the secondary component radius is calculated from multiplying the radius ratio k by the primary component ratio, reproducing a similarly skewed shape.

5.6. *Future work*

We have demonstrated the flexibility of our framework to infer all orbital parameters for a selection of spectroscopic eclipsing binaries with varied TESS light curve geometries and varying number of APOGEE radial velocity measurements. The next step will be to make modifications to the framework to increase its flexibility in modeling SEBs. These improvements will focus on for example, expanding the framework to handle eccentricities $\gg 0$, adding more options to the Gaussian process, such as different kernels to model the out-of-eclipse flux, and adding a treatment for SB2 radial velocity data into the orbital model. Constraints on the mass-radius parameter space may be implemented to avoid non-physical inferences. Afterward the framework will be used to model all identified binaries within the *The Joker* catalog using the available TESS photometry, and the APOGEE radial velocity measurements to construct a catalog of complete binary orbital parameters that have been homogeneously treated. A catalog of complete binary orbital parameters can be

Figure 5. Plot of the masses and radii for primary (blue diamonds) and secondary (orange diamonds) components in the 12 SEBs considered in this work. The 1σ contours (blue and orange lines) for each binary component's mass and radius posterior distributions are plotted to highlight the covariance between mass and radius. We plot the mass-radius measurements from Torres et al. (2010b) for the primary (filled black circles) and secondary (empty black circles) components. The vertical/horizontal gray dashed line demarcate the minimum determined mass/radius from the Torres et al. (2010b) catalog, respectively. We plot mass-radius values of a solar metallicity MIST isochrone at 5 Myr (solid red line), 100 Myr (dash red line), 1000 Myr (dashed dot red line), and 12000 Myr (dotted red line).

Figure 6. Plot of three separate orbital parameters as well as stellar parameters one to one plots comparing the inferred values for our binary systems and the values derived by the *Gaia* collaboration from their recently released *Gaia* DR3 (Holl et al. 2022) as well as APOGEE(Majewski et al. 2017). Panel A is comparing the orbital period of the binary system. Panel B is a plot comparing the orbital eccentricity for our binary sample. Panel C is a stellar parameter comparison plot of the inferred surface gravity, $\log g$, of the primary stars in our sample with the $\log g$ values calculated by APOGEE.

used to perform hierarchical Bayesian modeling of important distributions of the Milky Way binary population.

6. CONCLUSION

In this work we have introduced a novel framework to infer orbital parameters for 12 spectroscopic eclipsing binaries (SEBs). We use available photometric TESS data and the spectroscopic APOGEE data together to infer the parameters of a two-component binary orbital model within a Bayesian Probabilistic framework.

The TESS light curves are downloaded using the LightKurve package and a high resolution box-least squares (BLS) fit is performed to infer an initial orbital period and time of eclipse. The TESS light phase curve and the APOGEE radial velocity phase curve are centered and folded by this initial orbital period and time of eclipse for 11 out of the 12 binary systems. We find the two similar parameters derived using only radial velocities within the *The Joker* catalog (Price-Whelan et al. 2020) resulted in a better folded phase curve for TIC 169820068.

This framework uses Hamiltonian Monte Carlo (HMC) to perform sampling of posteriors. We select HMC to perform sampling within our framework because of its reliability to properly explore a parameter space, as well as scaling well to large number of dimensions. Our framework is constructed to be able to perform all necessary operations to model an binary system without further manual modification because of the built in capabilities of Exoplanet that we construct our framework upon.

We run our framework around a small sample of 12 spectroscopically eclipsing binaries. We use the Isochrones package to derive informed values for the parameters that we use to initialize our priors for each binary orbital model. We recover reliable SEB model parameter distributions for 11 of our binary systems using the same framework with almost no manual intervention necessary. Only one of our systems has a partially reliable model parameter inference, due to a non-physical secopndary radius. We will continue this work and scaling our framework to fit the larger sample of binaries within both the APOGEE and TESS surveys in order to construct a benchmark catalog of SEBs with fully modeled orbital parameters.

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Software: Arviz (Kumar et al. 2019), Astropy (Astropy Collaboration et al. 2013, 2018, 2022), Celerite2 (Foreman-Mackey 2018), Exoplanet (Foreman-Mackey et al. 2021), IPython (Perez & Granger 2007), Isochrones (Morton 2015), LightKurve (Cardoso & Maselli 2019), matplotlib, a Python library for publication quality graphics (Hunter 2007), NumPy (Harris et al. 2020), Pandas (McKinney 2010, 2011) PyMC3 (Salvatier et al. 2016), Scipy (Virtanen et al. 2020), Theano (Al-Rfou et al. 2016)

APPENDIX

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Figure 7. Corner plot of posterior values of ($\log M_1$, $\log R_1$, $\log q$, $\log s$, b, t_0, t_N, ecc, ω) for binary system TIC 196820068. The 2-dimensional contour on each corner is the equivalent 1-σ. The orange lines on each contour plot are the final MAP values for each variable after the 2nd round of MAP optimizations (see Section 3.6). The left and right vertical dashed blue lines on each 1-dimensional histogram plots of each variable are the 16th and 84th quantiles, respectively. **Fig. Set 7. PyMC3 model posterior corners**

Figure 8. Side by side plots of all light curves (left panel) and all radial velocity curves (right panel). Both the radial velocity data and the light curve data have been plotted chronologically. The orbital model constructed with the best parameters are used to produce both the model eclipse light curve (left panel, orange line) and the radial velocity curve (right panel, orange line). Light curve data and radial velocity data are plotted in both panels as black dots. Each system also has its full orbital period is plotted as blue shading. One full orbital period is plotted for each plot starting from either the first eclipse or the start of radial velocity observations.

Figure 9. Residuals between model generated observations and real observations for TIC-20215452 (a), TIC-28159019 (b), TIC-99254945 (c), TIC-126232983 (d), TIC-144441148 (e), and TIC-164458426 (f). The top panel for each system are the residuals between model generated light curve flux and the TESS observations light curve flux in ppt. The black solid line in the top panel is a rolling median of the light curve residuals for each system. The bottom panel for each system are the residuals between model generated radial velocity measurements and the APOGEE radial velocity observations in $\mathrm{km\,s^{-1}}$. The residuals are folded with the primary epoch time t_0 and plotted to lie between $\pm 0.5 \times P_{orbital}$.

Figure 10. Residuals between model generated observations and real observations for TIC-164527723 (g), TIC-165453878 (h), TIC-169820068 (i), TIC-258108067 (j), TIC-271548206 (k), and TIC-272074664 (l). Plot and panel layouts are the same as Figure 9

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